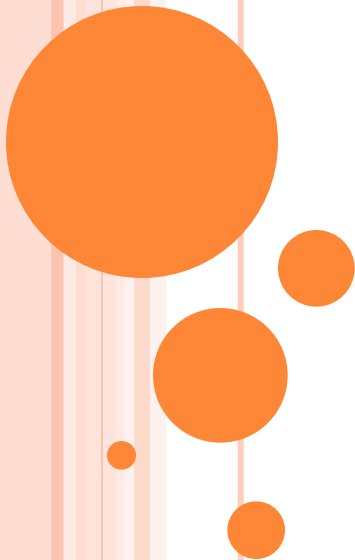


PROPOSITIONAL LOGIC

Syntactic Approach



AXIOMATIC (FORMAL) SYSTEM: $P = (\Sigma_P, F_P, A_P, R_P)$

- $\Sigma_P = Var_propos \cup Connectives \cup \{ (,) \}$ - vocabulary
 $Var_propos = \{ p_1, p_2, \dots \}$ - a *set of propositional variables*
 $Connectives = \{ \neg, \wedge, \vee, \rightarrow, \leftrightarrow, \oplus, \uparrow, \downarrow \};$
- $F_P =$ *the set of well formed formulas,*
- $A_P = \{ A1, A2, A3 \}$ - the *set of axioms* |
 $A1: U \rightarrow (V \rightarrow U)$
 $A2: ((U \rightarrow (V \rightarrow Z)) \rightarrow ((U \rightarrow V) \rightarrow (U \rightarrow Z)))$
 $A3: (U \rightarrow V) \rightarrow (\neg V \rightarrow \neg U)$ (modus tollens)
- $R_P = \{ mp \}$ - the *set of inference (deduction) rules* containing *modus ponens rule*.

notation: $U, U \rightarrow V \vdash_{mp} V$

with the meaning: “from the facts U and $U \rightarrow V$ we *deduce (infer)* V ”.

DEDUCTION

Definition: Let $U_1, U_2, \dots, U_n$ be propositional formulas, called *hypotheses* and V be a formula, called *conclusion*. V is **deducible (derivable, inferable)** from $U_1, \dots, U_n$ and we denote by $U_1, \dots, U_n \vdash V$, if there exists a sequence $(f_1, f_2, \dots, f_m)$ of formulas such that $f_m = V$ and $\forall i \in \{1, \dots, m\}$ we have a) or b) or c).

- a) $f_i \in A_P$ (axiom);
- b) $f_i \in \{U_1, \dots, U_n\}$ (hypothesis formula);
- c) $f_{i1}, f_{i2} \vdash_{mp} f_i, \quad i1 < i \text{ and } i2 < i$

(formula f_i is inferred using *modus ponens* rule from two existing formulas)

The sequence $(f_1, f_2, \dots, f_m)$ is called the *deduction of* V from $U_1, U_2, \dots, U_n$.

Definition: A formula $U \in F_P$, such that $\emptyset \vdash U$ (or $\vdash U$) is called *theorem*.

Remark: The theorems are the formulas derivable (inferable) only from the axioms and using modus ponens as inference rule.



PROPERTIES OF DERIVABILITY RELATION

Theorem:

Let R, S be sets of propositional formulas and U, V, Z , be formulas.
The derivability (syntactic consequence) relation has the properties:

1. *monotonicity:*

if $R \vdash U$ and $R \subseteq S$ then $S \vdash U$;

2. *cut:*

if $S \vdash V_j$, $\forall j \in \{1, \dots, n\}, n \in \mathbf{N}$ and $S \cup \{V_1, V_2, \dots, V_n\} \vdash U$ then $S \vdash U$;

3. *transitivity:*

if $S \vdash U$ and $\{U\} \vdash V$ then $S \vdash V$;

4. *conjunction in conclusions (right "and"):*

if $S \vdash U$ and $S \vdash V$ then $S \vdash U \wedge V$;

5. *disjunction in premises (left "or"):*

if $S \cup \{U\} \vdash Z$ and $S \cup \{V\} \vdash Z$ then $S \cup \{U \vee V\} \vdash Z$;



EXAMPLE 1 OF REASONING MODELING: *PARTY*

- H1:** Mary will go to the party if Lucy will go and George will not go.
- H2:** If John will go to the party then Lucy will go too.
- H3:** If John is in town he will go to the party.
- H4:** George is sick and can't go to the party.
- H5:** Yesterday John has returned in town from Paris.
- C(onclusion):** Will Mary go to the party?

We have to check whether the following deduction holds.
 $H1, H2, H3, H4, H5 \vdash C$

Notations for the propositional variables:

- M – Mary will go to the party
 L – Lucy will go to the party
 G – George will go to the party
 J – John will go to the party
 Jt – John is in town

Propositional formulas:

- H1:** $L \wedge \neg G \rightarrow M$
H2: $J \rightarrow L$
H3: $Jt \rightarrow J$
H4: $\neg G$
H5: Jt
C: M



EXAMPLE 1 - BUILDING THE DEDUCTION



M.Lupea

The definition of the deduction and the axiomatic system are used.

f1=H1: $L \wedge \neg G \rightarrow M$ (hypothesis)

f2=H2: $J \rightarrow L$ (hypothesis)

f3=H3: $Jt \rightarrow J$ (hypothesis)

f4=H4: $\neg G$ (hypothesis)

f5=H5: Jt (hypothesis)

$$U, U \rightarrow V \vdash_{mp} V$$

The deduction (inference) process:

$f5, f3 \vdash_{mp} J : \mathbf{f6}$ (*modus ponens is applied*)

$f6, f2 \vdash_{mp} L : \mathbf{f7}$ (*modus ponens is applied*)

$f4, f7 \vdash L \wedge \neg G : \mathbf{f8}$ (*conjunction in conclusions*)

$f8, f1 \vdash_{mp} M : \mathbf{f9} = \mathbf{C}$ (*modus ponens is applied*)

The sequence of formulas: (f1, f2, f3, f4, f5, f6, f7, f8, f9) is the *deduction* of C from the hypotheses therefore, based on the hypotheses , **Mary will go to the party.**



EXAMPLE 2

Prove that: $\neg p \vee q, p \vee r, \neg q \vdash r$ holds using the **definition of deduction**.

We build the sequence (f1,f2,f3, f4, f5,f6,f7) of formulas as follows:

$$f1: \neg p \vee q \equiv p \rightarrow q$$

$$U, U \rightarrow V \vdash_{mp} V$$

$$f2: p \vee r \equiv \neg p \rightarrow r$$

$$U \rightarrow V \equiv \neg U \vee V$$

$$f3: \neg q$$

$$f4: (p \rightarrow q) \rightarrow (\neg q \rightarrow \neg p) \text{ - axiom A3 (modus tollens)}$$

$$f1, f4 \vdash_{mp} \neg q \rightarrow \neg p$$

$$f5: \neg q \rightarrow \neg p$$

$$f3, f5 \vdash_{mp} \neg p$$

$$f6: \neg p$$

$$f2, f6 \vdash_{mp} r$$

$$f7: r$$

$f7 = r$ was proved from the hypotheses (f1,f2,f3) using modus ponens and A3.

PROPERTIES OF PROPOSITIONAL LOGIC

Soundness theorem (*syntactic validity implies semantic validity*):

If $\vdash U$ then $\models U$ (A theorem is a tautology).

Completeness theorem (*semantic validity implies syntactic validity*):

If $\models U$ then $\vdash U$ (A tautology is a theorem).

Theorem of soundness and completeness for propositional logic:

$\vdash U$ if and only if $\models U$.

Consequences of this theorem are the following **properties**:

- **Propositional logic is *non-contradictory***: we can't have simultaneously $\vdash U$ and $\vdash \neg U$.
- **Propositional logic is *coherent***: not every propositional formula is a theorem.
- **Propositional logic is *decidable***: we can always decide whether a propositional formula is a theorem or not. The truth table method is a decision method.



PROPOSITIONAL INFERENCE RULES

	Inference rule	Theorem
<i>Addition</i>	$U \vdash U \vee V$	$\vdash U \rightarrow U \vee V$
<i>Simplification</i>	$U \wedge V \vdash U$	$\vdash U \wedge V \rightarrow U$
	$U \wedge V \vdash V$	$\vdash U \wedge V \rightarrow V$
<i>Modus ponens</i>	$U, U \rightarrow V \vdash V$	$\vdash U \wedge (U \rightarrow V) \rightarrow V$
<i>Modus tollens</i>	$U \rightarrow V \vdash \neg V \rightarrow \neg U$	$\vdash (U \rightarrow V) \rightarrow (\neg V \rightarrow \neg U)$
	$\neg V, U \rightarrow V \vdash \neg U$	$\vdash \neg V \wedge (U \rightarrow V) \rightarrow \neg U$
<i>Syllogism</i>	$U \rightarrow V, V \rightarrow Z \vdash U \rightarrow Z$	$\vdash (U \rightarrow V) \wedge (V \rightarrow Z) \rightarrow (U \rightarrow Z)$
	$U, U \rightarrow V, V \rightarrow Z \vdash Z$	$\vdash U \wedge (U \rightarrow V) \wedge (V \rightarrow Z) \rightarrow Z$
<i>Resolution</i>	$U \vee V, \neg U \vee Z \vdash V \vee Z$	$\vdash (U \vee V) \wedge (\neg U \vee Z) \rightarrow (V \vee Z)$



Modus ponens

$$p, p \rightarrow q \vdash q$$

	p	q	$p \rightarrow q$
i1	T	T	T
i2	T	F	F
i3	F	T	T
i4	F	F	T

Modus ponens - Latin

“mode that by affirming the premise (p) affirms the conclusion (q)”

Modus tollens

$$\neg q, p \rightarrow q \vdash \neg p$$

Modus tollens - Latin

“mode that by denying the conclusion (q) denies the premise (p)”



WHICH RULE OF INFERENCE IS USED IN EACH ARGUMENT BELOW?

H. Alice is a Math major. (p) C. Therefore, Alice is either a Math major or a CSI major. $(p \vee q)$	<i>Addition</i> $p \vdash p \vee q$
H. Jerry is a Math major and a CSI major $(p \wedge q)$ C. Therefore, Jerry is a Math major. (p)	<i>Simplification</i> $p \wedge q \vdash p$
H1. If it is rainy, then the pool will be closed. $(p \rightarrow q)$ H2. It is rainy. (p) C. Therefore, the pool is closed. (q)	<i>Modus ponens</i> $p, p \rightarrow q \vdash q$
H1. If it snows today, the university will close $(p \rightarrow q)$ H2. The university is not closed today. $(\neg q)$ C. Therefore, it did not snow today. $(\neg p)$	<i>Modus tollens</i> $\neg q, p \rightarrow q \vdash \neg p$
H1. If I go swimming, then I will stay in the sun too long $(p \rightarrow q)$ H2. If I stay in the sun too long, then I will sunburn. $(q \rightarrow r)$ C. Therefore, if I go swimming, then I will sunburn. $(p \rightarrow r)$	<i>Syllogism</i> $p \rightarrow q, q \rightarrow r \vdash p \rightarrow r$
H1. I go swimming or eat an ice cream $(p \vee q)$ H2. I do not go swimming $(\neg p)$ C. Therefore, I eat an ice cream (q)	<i>Resolution</i> $p \vee q, \neg p \vdash q$

EXAMPLE 3 OF REASONING MODELING

Build a deduction of the conclusion (C) from the hypotheses H1, H2, H3, H4.

H1: If it is rainy Kate goes to the library or she goes shopping.

H2: Kate goes shopping **only with her friend Carla.**

H3: John goes to the library daily.

H4: It is rainy today, Carla is sick and she will not go shopping.

C (conclusion) Kate will meet John at the library today.



ÎNȚELEPCIUNE



Trei lucruri care nu se întorc niciodată înapoi:

1. Timpul;
2. Cuvântul;
3. Șansa

Trei lucruri pe care nu trebuie să le pierzi:

1. Calmul;
2. Cinstea;
3. Speranța

Trei lucruri sunt valoroase cu adevărat:

1. Dragostea;
2. Respectul;
3. Încrederea

Trei lucruri în viață nu sunt sigure:

1. Succesul;
2. Puterea;
3. Averele

Trei lucruri care îl caracterizează pe om:

1. Munca;
2. Corectitudinea;
3. Performanța

Trei lucruri care îl distrug pe om:

1. Băutura;
2. Aroganța;
3. Răutatea

Trei lucruri care sunt mai greu de spus:

1. Te iubesc!
2. Ajută-mă!
3. Iartă-mă!

FALLACIES

	p	q	$p \rightarrow q$
i1	T	T	T
i2	T	F	F
i3	F	T	T
i4	F	F	T

Fallacy: mistaken belief based on unsound argument.

- **Fallacy of affirming the conclusion**

H1. If it is Saturday I go to the swimming pool. ($p \rightarrow q$)

H2. I go to the swimming pool. (q)

C. Is it Saturday? (Can we infer p ?)

$p \rightarrow q, q \not\models p$ because $\models (p \rightarrow q) \wedge q \rightarrow p$

- **Fallacy of denying the hypothesis**

H1. If it is Saturday I go to the swimming pool. ($p \rightarrow q$)

H2. It is **not** Saturday. ($\neg p$)

C. I do **not** go to the swimming pool. (Can we infer ($\neg q$)?)

$p \rightarrow q, \neg p \not\models \neg q$ because $\models (p \rightarrow q) \wedge \neg p \rightarrow \neg q$



FALLACIES (contd.)

- Fallacy of affirming a disjunct

H1: I am at home or I am in the city. ($p \vee q$)

H2: I am at home. (p)

C: I am **not** in the city. (Can we infer ($\neg q$)?)

$p \vee q, p \not\models \neg q$ because $\not\models (p \vee q) \wedge p \rightarrow \neg q$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

- Fallacy of denying a conjunct

H1: I **cannot** be both at home and in the city. ($\neg(p \wedge q)$)

H2: I am **not** at home. ($\neg p$)

C: I am in the city. (Can we infer (q)?)

$\neg(p \wedge q), \neg p \not\models q$ because $\not\models \neg(p \wedge q) \wedge \neg p \rightarrow q$



THEOREM OF DEDUCTION AND ITS REVERSE

Theorem of deduction:

If $U_1, \dots, U_{n-1}, U_n \vdash V$, then $U_1, \dots, U_{n-1} \vdash U_n \rightarrow V$.

Reverse of the theorem of deduction:

If $U_1, \dots, U_{n-1} \vdash U_n \rightarrow V$ then $U_1, \dots, U_{n-1}, U_n \vdash V$.

By applying “ n ” times the theorem of deduction and its reverse we obtain:

$U_1, \dots, U_{n-1}, U_n \vdash V$ **if and only if**

$U_1, \dots, U_{n-1} \vdash U_n \rightarrow V$ **if and only if**

$U_1, \dots, U_{n-2} \vdash U_{n-1} \rightarrow (U_n \rightarrow V)$ **if and only if**

...

$U_1 \vdash U_2 \rightarrow (\dots U_{n-1} \rightarrow (U_n \rightarrow V) \dots)$ **if and only if**

$\vdash U_1 \rightarrow (U_2 \rightarrow (\dots \rightarrow (U_n \rightarrow V) \dots))$



Theorem of deduction - consequences

1. $\vdash U \rightarrow ((U \rightarrow V) \rightarrow V)$
2. $\vdash (U \rightarrow V) \rightarrow ((V \rightarrow Z) \rightarrow (U \rightarrow Z))$
3. $\vdash (U \rightarrow (V \rightarrow Z)) \rightarrow (V \rightarrow (U \rightarrow Z))$ *permutation of the premises law*
4. $\vdash (U \rightarrow (V \rightarrow Z)) \rightarrow (V \wedge U \rightarrow Z)$ *reunion of the premises law*
5. $\vdash (U \wedge V \rightarrow Z) \rightarrow (U \rightarrow (V \rightarrow Z))$ *separation of the premises law*



PROOF

$$2. \quad \vdash (U \rightarrow V) \rightarrow ((V \rightarrow Z) \rightarrow (U \rightarrow Z))$$

We begin with the deduction:

$$U, U \rightarrow V, V \rightarrow Z \vdash Z \text{ (syllogism)}$$

Application of the theorem of deduction $\Rightarrow U \rightarrow V, V \rightarrow Z \vdash U \rightarrow Z$

Application of the theorem of deduction $\Rightarrow$

$$U \rightarrow V \vdash (V \rightarrow Z) \rightarrow (U \rightarrow Z)$$

Application of the theorem of deduction $\Rightarrow$

$$\vdash (U \rightarrow V) \rightarrow ((V \rightarrow Z) \rightarrow (U \rightarrow Z))$$



EXAMPLE 4 – THEOREM OF DEDUCTION

Using the theorem of deduction and its reverse prove:

$$\vdash (p \rightarrow r) \rightarrow ((p \wedge r \rightarrow q) \rightarrow (p \rightarrow q))$$

Step1: We apply the **reverse of the theorem of deduction** to obtain
the initial deduction.

The premise of the main implication is moved from right to the left.

if $\vdash (p \rightarrow r) \rightarrow ((p \wedge r \rightarrow q) \rightarrow (p \rightarrow q))$ **then**

$p \rightarrow r \vdash (p \wedge r \rightarrow q) \rightarrow (p \rightarrow q)$ **then**

$p \rightarrow r, p \wedge r \rightarrow q \vdash p \rightarrow q$ **then**

$p \rightarrow r, p \wedge r \rightarrow q, p \vdash q$



EXAMPLE 4 (CONTD.)

Step2: We prove the deduction obtained at Step1:

$$p \rightarrow r, p \wedge r \rightarrow q, p \vdash q ,$$

building the sequence of formulas: (f1, f2, f3, f4, f5, f6):

f1: p --- premise (hypothesis)

f2: $p \rightarrow r$ --- premise

$$f1, f2 \vdash_{mp} r$$

f3: r

f4: $f1 \wedge f3 = p \wedge r$ (conjunction of the conclusions)

f5: $p \wedge r \rightarrow q$ --- premise

$$f4, f5 \vdash_{mp} q$$

f6: q

The sequence (f1, f2, f3, f4, f5, f6) is the deduction of q from the premises $p \rightarrow r, p \wedge r \rightarrow q, p$.



EXAMPLE 4 (CONTD.)

Step3: We begin with the deduction $p \rightarrow r, p \wedge r \rightarrow q, p \vdash q$ proved at **Step2** and we apply three times the theorem of deduction.

There are $3!=6$ such possibilities (to move the premises to the right side of the meta-symbol $\vdash$) and we prove 6 theorems: T1, T2, T3, T4, T5, T6.

The premises are moved to the right-hand side of ' $\vdash$ ' in the following order:

$$p, p \wedge r \rightarrow q, p \rightarrow r$$

if $p \rightarrow r, p \wedge r \rightarrow q, p \vdash q$ **then**

$p \rightarrow r, p \wedge r \rightarrow q \vdash p \rightarrow q$ **then**

$p \rightarrow r \vdash (p \wedge r \rightarrow q) \rightarrow (p \rightarrow q)$ **then**

$\vdash T1 = (p \rightarrow r) \rightarrow ((p \wedge r \rightarrow q) \rightarrow (p \rightarrow q))$ --- the theorem to be proved.

The following theorems can be also proved:

$$\vdash T2 = (p \wedge r \rightarrow q) \rightarrow ((p \rightarrow r) \rightarrow (p \rightarrow q))$$

$$\vdash T3 = (p \wedge r \rightarrow q) \rightarrow (p \rightarrow ((p \rightarrow r) \rightarrow q))$$

$$\vdash T4 = p \rightarrow ((p \wedge r \rightarrow q) \rightarrow ((p \rightarrow r) \rightarrow q))$$

$$\vdash T5 = (p \rightarrow r) \rightarrow (p \rightarrow ((p \wedge r \rightarrow q) \rightarrow q))$$

$$\vdash T6 = p \rightarrow ((p \rightarrow r) \rightarrow ((p \wedge r \rightarrow q) \rightarrow q))$$



DECISION PROBLEMS IN PROPOSITIONAL LOGIC

1. Is a propositional formula a tautology/theorem)?

$$\overset{?}{\models} V \quad \text{or} \quad \overset{?}{\vdash} V$$

2. Is a propositional formula a logical/syntactic consequence of a set of hypotheses?

$$U_1, \dots, U_n \overset{?}{\models} V \quad \text{or} \quad U_1, \dots, U_n \overset{?}{\vdash} V$$

In order to solve these two decision problems, **theorem proving methods** are applied.



CLASSIFICATION OF PROOF METHODS

semantic <i>versus</i> syntactic methods	direct <i>versus</i> refutation methods
<u><i>semantic methods</i></u> • the truth table method; • the semantic tableaux method; • the CNF- conjunctive normal form.	<u><i>direct methods</i></u>: they use directly the formula to be proved • the truth table method; • the CNF- conjunctive normal form; • the definition of deduction; • the theorem of deduction and its reverse; • the sequent calculus method.
<u><i>syntactic methods</i></u> • the definition of deduction; • the theorem of deduction and its reverse; • the resolution method; • the sequent calculus method.	<u><i>refutation methods</i></u>: they model the “reductio ad absurdum” (proof by contradiction) and they use the negation of the formula to be proved • the semantic tableaux method; • the resolution method.